MTH 310 **Quiz 4** Name:____________________

Spring, 2026

Instructions. (15 points) Sections 14.4–14.5. Show relevant work for full credit.

(8pts) **1.** A grain silo consists of a cylinder of radius $r = 3$ m and height $h = 8$ m topped by a hemisphere of radius $r = 3$ m. The silo is built from wooden planks that are 0.05 m thick.

(a) *(4 pts)* The volume of the silo is $V = \pi r^2 h + \frac{2}{3}\pi r^3$. Find the total differential dV.

(b) *(4 pts)* Determine the appropriate values of dr and dh for this problem, then use differentials to estimate the total volume of wood in the silo walls, roof, and floor.

8 pts

(7pts) **2.** Let $w = xy + yz + xz$, where $x = s^2 + t$, $y = st$, and $z = t^2$.

(a) *(3 pts)* Draw the tree diagram and write the chain rule expression for $\dfrac{\partial w}{\partial t}$.

(b) *(4 pts)* Evaluate $\dfrac{\partial w}{\partial t}$ at $(s, t) = (1, 2)$.

7 pts